

Algorithm

Step-by-Step Procedure

1. Collect the dataset from a reliable source.
2. Load the dataset into the programming environment.
3. Preprocess the data by handling missing values, duplicates, and encoding categorical variables if required.
4. Split the dataset into training and testing sets.
5. Train the selected machine learning model using the training data.
6. Test the trained model using the testing data.
7. Evaluate the model using performance metrics such as accuracy, precision, recall, and F1-score.
8. Use the trained model to make predictions on new input data.

Logic or Method Used

The system learns patterns from historical data using a supervised machine learning algorithm. During training, the model identifies relationships between the input features and the target variable. After training, it predicts the output for unseen data.

Model or Technique Applied

(Replace this with the model used in your project.)

Examples:

- Decision Tree
- Random Forest
- Logistic Regression
- Support Vector Machine (SVM)
- Naïve Bayes
- K-Nearest Neighbors (KNN)
- Neural Network

Example:

Model Used: Random Forest Classifier

Process Flow of the System

Dataset Collection



Data Preprocessing



Feature Selection



Train-Test Split



Model Training



Model Testing



Performance Evaluation



Prediction

Dataset

Source of Data

Mention where the dataset was obtained.

Examples:

- Kaggle
- UCI Machine Learning Repository
- Government Open Data Portal
- Hospital/Company Database
- Self-collected data

Example:

Source: Kaggle Dataset

Number of Records/Samples

Example:

- Total Records: 5,000
- Training Records: 4,000
- Testing Records: 1,000

(Replace these numbers with your dataset details.)

Features or Attributes (Columns)

Example:

Feature	Description
Age	Age of the person
Gender	Male/Female
Blood Pressure	Blood pressure level
Cholesterol	Cholesterol level
Heart Rate	Heart rate
Target	Disease prediction (Yes/No)

Replace these with the actual columns in your dataset.

Training and Testing Data Split

A common split is:

- Training Data: **80%**
- Testing Data: **20%**

or

- Training Data: **70%**
- Testing Data: **30%**

Example:

Total Samples : 5000

Training Data : 4000 (80%)

Testing Data : 1000 (20%)

Data Format or Type

Example:

- File Format: CSV (.csv)
- Data Type: Structured tabular data
- Features: Numerical and categorical
- Target Variable: Binary or multiclass classification (depending on the project)

Example Summary

Item	Details
Dataset Source	Kaggle
Total Samples	5,000
Number of Features	12
Target Variable	Disease Prediction
Training Split	80%
Testing Split	20%
Data Format	CSV
Model Used	Random Forest Classifier